INSTITUTE OF MATHEMATICAL SCIENCES, FACULTY OF SCIENCE, UNIVERSITI MALAYA
SEMESTER 2, SESSION 2025/2026
TUTORIAL 5

1. A tank is filled with 10 gallons of brine in which $30\ lb$ of salt is dissolved. Brine having $2\ lb$ of salt per gallon enters the tank at a rate of 2 gal per min and the well- stirred mixture leaves at a rate of 3 gal per min.
    a. Set up a differential equation for the amount of salt at time t.
    b. Find the amount of salt and the concentration of salt after 5 minutes.
    c. After 5 minutes, the rate of brine entering the tank is increased to 3 gallons per min. Find the amount of salt in the tank after 10 minutes.

2. A big tank is filled with x_0 amount of salt which is dissolved in $100L$ water. A solution with ¼ kg of salt per litre the tank at the rate r litre per min. Well-stirred mixture leaves the tank at the rate of r litre per min.
    (a) Set up a differential equation with an initial value to describe the process.
    (b) Solve the equation and obtain the value of the limit x_L , as time $t \to \infty$.
    (c) If given that $r = 3$, and $x_0 = 2x_L$, determine the time when the salt is around 2% of x_L.
    (d) If as in part (c) but r is not given, obtain the rate r L/min such time T is not more than 45min.

3. A 1000-gallon tank is initially half filled with water in which 10 lb of salt is dissolved. Pure water enters the tank at the rate of 6 gal/min. The well-stirred mixture leaves the tank at the rate of 1 gal/min. Determine the concentration of salt in the tank as a function of time until the tank is full. How long will it take to fill up the tank?

4. Initially a tank is filled up with 150 gallons of water in which $300\ lb$ of salt is dissolved. Brine with concentration of 1 lb/gal enters the tank at the rate of 5 gal/min. The solution in the tank is kept well-stirred and flows out of the tank at the rate of 3 gal/min.
    (a) Find the minimum amount of salt in the tank.
    (b) Find the time taken to reach the minimum amount.

5. Tank A contains 100 gallons of brine in which 50 lb of salt is dissolved. Pure water enters at a rate of 2 gal/min. The well-stirred mixture flows out at the same rate into tank B which initially contains 100 gallons of pure water. The well-stirred mixture flows out of tank B at a rate of 2 gal/min.
    a. If A and B contain x lb and y lb of salt respectively, express x and y as a function of t.
    b. Obtain the amount of salt in tank B after 1 hour.